

# Quantitative Research Methods

## Conformal Prediction

Advanced module A3 — optional self-study

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# Where this module fits

The four **advanced modules** are optional self-study decks that extend the twelve ISLP lectures. This one turns any predictive model from the course into a model with *honest uncertainty*.

| You already know (chapter)                         | What module A3 adds                                                                   |
|----------------------------------------------------|---------------------------------------------------------------------------------------|
| Ch. 3 — prediction intervals for linear regression | Intervals with <i>guaranteed</i> coverage, without Gaussian errors or a correct model |
| Ch. 5 — train / validation splits                  | A third role for held-out data: the <i>calibration set</i>                            |
| Ch. 4 — classifiers and their probabilities        | Prediction sets: “{flu, covid}, with 90% guarantee”                                   |
| Ch. 8 — gradient boosting                          | Boosted <i>quantile</i> regression as the engine of adaptive intervals (CQR)          |

Prerequisites: Chapters 3 and 5 are essential, Chapters 4 and 8 helpful. Short exercises every  $\sim 20$  min; extended exercises every  $\sim 45$  min; the module has a companion lab notebook.

# Contents

- 1 Why Conformal
- 2 Split Conformal
- 3 Adaptive Intervals
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Optional and advanced material is collected in the **appendix** at the end of the deck.

# Notation in this module

| Symbol                                              | Meaning                                                                  |
|-----------------------------------------------------|--------------------------------------------------------------------------|
| $(x_i, y_i)$                                        | feature–response pair of observation $i$                                 |
| $\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{cal}}$ | proper training set and calibration set (disjoint)                       |
| $n$                                                 | number of calibration points, $n =  \mathcal{D}_{\text{cal}} $           |
| $\hat{f}$                                           | prediction rule fitted on $\mathcal{D}_{\text{tr}}$ only                 |
| $s_i$                                               | conformity score of calibration point $i$ (e.g. $ y_i - \hat{f}(x_i) $ ) |
| $s_{(k)}$                                           | $k$ -th smallest calibration score (order statistic)                     |
| $\alpha$                                            | target miscoverage level (we use $\alpha = 0.1$ : 90% coverage)          |
| $\hat{q}$                                           | conformal quantile: $s_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil$   |
| $C(x)$                                              | prediction interval (regression) or prediction set (classification)      |
| $\rho_\tau(u)$                                      | pinball loss of quantile regression at level $\tau$                      |
| $\hat{q}_{\text{lo}}, \hat{q}_{\text{hi}}$          | fitted conditional quantile curves (levels $\alpha/2, 1 - \alpha/2$ )    |
| $\hat{p}_k(x)$                                      | estimated probability of class $k$ at $x$ (softmax output)               |
| $K$                                                 | number of classes                                                        |

# Sources and references

## Primary textbook

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python*. Springer. (Chapters 3 and 5 supply the prerequisites.)

## Further reading on conformal prediction

Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer. — Lei, J., et al. (2018). Distribution-free predictive inference for regression. *JASA* 113(523). — Romano, Y., Patterson, E., & Candès, E. (2019). Conformalized quantile regression. *NeurIPS*. — Angelopoulos, A., & Bates, S. (2023). Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning* 16(4).

# Learning objectives

By the end of this module you will be able to:

- **Explain** why classical prediction intervals need assumptions that conformal prediction drops.
- **State** exchangeability and **judge** when it is plausible.
- **Apply** split conformal prediction: compute  $\hat{q}$  and the interval by hand and in Python.
- **Distinguish** marginal from conditional coverage — and say which one is guaranteed.
- **Build** adaptive intervals with conformalized quantile regression (CQR).
- **Construct** prediction sets for classifiers and read their sizes.
- **Avoid** the pitfalls: leakage into the calibration set, time series, tiny calibration sets.

# Roadmap of this module

## Uncertainty you can guarantee

- ① Why a point prediction is a guess — and what Ch. 3's intervals assume.
- ② Exchangeability: the one assumption conformal prediction needs.
- ③ Split conformal for regression: the whole recipe, and its proof sketch.
- ④ Marginal vs conditional coverage — the honest reading of the guarantee.
- ⑤ Adaptive intervals: locally weighted scores and CQR.
- ⑥ Prediction sets for classification; conformal vs the classical interval; pitfalls and production practice.

# Where conformal prediction is used in industry

| Sector                | Concrete application                                                                 | Which decision it drives                       |
|-----------------------|--------------------------------------------------------------------------------------|------------------------------------------------|
| Energy                | Intervals around day-ahead load and wind forecasts, recalibrated on recent residuals | How many MW of reserve capacity to buy         |
| E-commerce, logistics | Delivery-time windows behind the check-out promise                                   | Which arrival date is displayed and guaranteed |
| Pharma R&D            | Conformal sets in virtual screening of compounds                                     | Which molecules advance to the wet lab         |

## All three are paid for the width of the interval rather than the point forecast

In every row the model's *point* prediction was already available and already good; what was missing is a number that sizes the buffer, the promise or the shortlist. Conformal prediction supplies it **without trusting the model**: the coverage holds in finite samples for any  $\hat{f}$ , which is why a desk is willing to commit money to the width.



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# A point prediction without uncertainty is a guess

Take the Wage data ( $n = 3\,000$  workers): predict **wage** from **age**.

## How much do 45-year-olds actually earn?

Among the 268 workers aged 44–46, wages average \$120 400 — but they range from \$36 700 to \$318 300, with a standard deviation of \$43 400. A model that answers “\$120k” communicates *none* of that spread: the single number is a guess dressed up as knowledge.

- Every model from this course — linear regression, boosting, a neural net — produces  $\hat{f}(x)$ . The question this module answers: *how far can the truth be from  $\hat{f}(x)$ , with a guarantee?*
- We want a recipe that works for **any**  $\hat{f}$ , on **finite** samples, **without** trusting the model.

## Takeaway

A buyer sizing stock, a hospital planning beds, a grid operator buying reserve — none of them can act on “the forecast is 120”. They act on “between 80 and 160, 9 times out of 10”.

# What Chapter 3 gave us — and what it silently assumed

## The classical 90% prediction interval (Ch. 3)

Under the linear model  $Y = X\beta + \varepsilon$  with  $\varepsilon \sim \mathcal{N}(0, \sigma^2)$  i.i.d.:  $\hat{y}_0 \pm t_{n-p-1, 0.95} \hat{\sigma} \sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}$ .

### The fine print — three assumptions doing all the work

- ① **Correct mean model:**  $E[Y | X = x]$  really is linear in  $x$ .
- ② **Gaussian errors:** the  $t$ /normal quantile calibrates the width.
- ③ **Homoskedasticity:** one  $\sigma$  for every  $x$ .

Break any of the three and the “90%” printed by `statsmodels` is a claim, not a fact.

### Wage breaks the fine print

Wages are strongly right-skewed (skewness 1.68; 79 workers, 2.6%, earn above \$250k). The Gaussian recipe  $\hat{f}(x) \pm 1.645 \hat{\sigma}$  gives half-width 66.1 — and covers 95.2% of test wages while *claiming* 90%: miscalibrated and needlessly wide.

# The conformal promise

## What conformal prediction guarantees

From any fitted model  $\hat{f}$  and a calibration set of  $n$  held-out points, build  $C(x)$  such that for a new exchangeable point  $(X_{n+1}, Y_{n+1})$ :

$$\Pr(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

### Why this is remarkable — read the guarantee word by word

- **Distribution-free:** no Gaussian errors, no model for  $P(Y | X)$  — wages can be as skewed as they like.
- **Finite-sample:** holds exactly at  $n = 500$ , not just as  $n \rightarrow \infty$ .
- **Any model:**  $\hat{f}$  may be misspecified, overfit, or a black box.
- **Marginal:** averaged over the new point *and* the calibration draw — more on this honest word soon.

### Takeaway

Conformal trades “if my model is right, coverage is exact at every  $x$ ” for “*whatever* my model is, coverage is right *on average*”.

## Exercise A3.1 — What did we just buy? [Concept]

### Exercise

- 1 Name the three assumptions behind the Chapter 3 prediction interval.
- 2 Which of the three does conformal prediction keep, and what does it demand instead?
- 3 A colleague says: “With conformal prediction, my 90% interval covers the true wage of *every 45-year-old* with probability 90%.” Is that what the guarantee says?
- 4 Why does an overfit  $\hat{f}$  not break the conformal guarantee (hint: where are the residuals computed)?

# Solution — Exercise A3.1

## Solution

**Method.** Compare the two guarantees assumption by assumption.

- **Part 1.** Correct (linear) mean model; Gaussian errors; homoskedastic errors (one  $\sigma$  for all  $x$ ).
- **Part 2.** It keeps *none* of them. Instead it demands (i) **exchangeability** between calibration and test points and (ii) a **clean split**:  $\hat{f}$  never sees the calibration data.
- **Part 3.** No. The guarantee is **marginal**: averaged over all new points (and calibration draws), at least 90% are covered. Coverage *among 45-year-olds specifically* may be higher or lower — we will measure exactly this on Wage (98.6% for under-30s, 82.9% for 60-plus, same interval).
- **Part 4.** The scores are residuals on *held-out* calibration points. An overfit  $\hat{f}$  produces *large* held-out residuals, so  $\hat{q}$  grows and the interval honestly widens — the guarantee survives; only the width suffers.

## Common mistake

Reading “90%” as a per-person (conditional) probability. Marginal coverage is a statement about the *average* over the population, not about your specific  $x$ .

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# Exchangeability — the one assumption we need

## Definition

Random variables  $Z_1, \dots, Z_m$  are **exchangeable** if for every permutation  $\pi$ ,

$$(Z_{\pi(1)}, \dots, Z_{\pi(m)}) \stackrel{d}{=} (Z_1, \dots, Z_m).$$

## What it means in plain words

- Every *ordering* of the data is equally likely — the labels “calibration point 17” and “test point” carry no information.
- **i.i.d.**  $\Rightarrow$  **exchangeable**, but not conversely: exchangeability is *weaker* (e.g. sampling without replacement).
- Key consequence: among  $n + 1$  exchangeable scores with no ties, the **rank** of the last one is *uniform* on  $\{1, \dots, n + 1\}$ .

## Takeaway

Conformal prediction is rank arithmetic on exchangeable scores. Everything follows from “the new point is nothing special”.



# The honest slide: what breaks exchangeability

- **Time series:** today's demand depends on yesterday's — orderings are *not* equally likely, and tomorrow may differ systematically.
- **Covariate shift:** calibrate on last year's customers, deploy on a new market — the  $X$ -distribution moved.
- **Feedback loops:** the model's own decisions (credit granted, price shown) change the data it will be judged on.

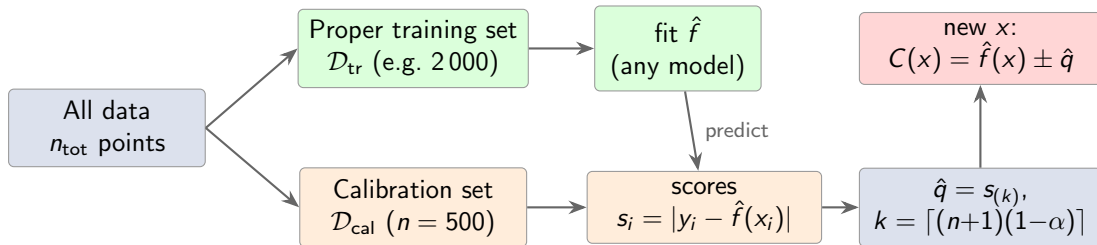
## What still works

Violations degrade the guarantee *gradually*, and there are repairs: *weighted* conformal under known covariate shift (Tibshirani et al., 2019), blocked or online variants for dependence. This module sticks to the exchangeable case — the pitfalls section returns here.

## Takeaway

Realised coverage is therefore also a *drift alarm*: sliding from 90% to 80% says the world has moved away from the calibration set, and it needs no labels beyond the ones you already collect.

# Split conformal — the pipeline (native diagram)



## Takeaway

One split, one model fit, one sorted list of residuals, one quantile. The model only ever sees  $\mathcal{D}_{\text{tr}}$ ; the guarantee only ever uses  $\mathcal{D}_{\text{cal}}$ .

# The whole recipe on one slide

## Algorithm: split conformal prediction for regression

- ① Split the data into  $\mathcal{D}_{\text{tr}}$  and  $\mathcal{D}_{\text{cal}}$  (sizes e.g. 2 000 and  $n = 500$ ).
- ② Fit  $\hat{f}$  on  $\mathcal{D}_{\text{tr}}$  *only*.
- ③ Compute calibration scores  $s_i = |y_i - \hat{f}(x_i)|$  for  $i \in \mathcal{D}_{\text{cal}}$ .
- ④ Set  $\hat{q} = s_{(k)}, \quad k = \lceil (n+1)(1-\alpha) \rceil$  — the  $k$ -th smallest score.
- ⑤ For any new  $x$  report  $C(x) = [\hat{f}(x) - \hat{q}, \hat{f}(x) + \hat{q}]$ .

## Decoding the boxed formula

$n$  = number of calibration points;  $\alpha$  = tolerated miscoverage (0.1 for a 90% interval);  $s_{(k)}$  = the  $k$ -th order statistic of the sorted scores; the ceiling  $\lceil \cdot \rceil$  rounds *up*, so  $\hat{q}$  is the  $\frac{k}{n}$  empirical quantile — slightly *above* the naive  $(1-\alpha)$ -quantile. With  $n = 500$ ,  $\alpha = 0.1$ :  $k = \lceil 501 \times 0.9 \rceil = 451$ , the 0.902 quantile.

# Why the odd index? A tiny worked example

## Nine calibration scores at 10% miscoverage

Scores: 1.2, 0.4, 2.8, 0.9, 1.7, 3.5, 0.6, 2.1, 1.0. Sorted: 0.4, 0.6, 0.9, 1.0, 1.2, 1.7, 2.1, 2.8, 3.5. Then  $k = \lceil (9 + 1)(0.9) \rceil = 9 \implies \hat{q} = s_{(9)} = 3.5$  (the maximum!). At  $\alpha = 0.2$ :  $k = \lceil 10 \times 0.8 \rceil = 8$ , so  $\hat{q} = s_{(8)} = 2.8$ . At  $n = 19$ ,  $\alpha = 0.1$ :  $k = \lceil 20 \times 0.9 \rceil = 18$  — the 18th of 19.

## The intuition for the “+1”

The new point's score joins the  $n$  calibration scores as an  $(n+1)$ -th, equally-ranked member. To catch it with probability  $\geq 0.9$  we must clear at least 90% of *all*  $n + 1$  ranks — hence  $(n + 1)(1 - \alpha)$ , rounded up because ranks are integers. The naive 90% quantile of  $n$  scores would undercover, worst at small  $n$ .

## Takeaway

Small calibration sets force conservative quantiles: with  $n = 9$  the 90% interval must stretch to the *largest* observed residual.

# The coverage guarantee, and why it is true

## Theorem (split conformal coverage)

If the calibration pairs and the test pair are exchangeable, and  $\hat{f}$  was fit without touching them, then

$$1 - \alpha \leq \Pr(Y_{n+1} \in C(X_{n+1})) \leq 1 - \alpha + \frac{1}{n+1},$$

the upper bound requiring almost-surely distinct scores.

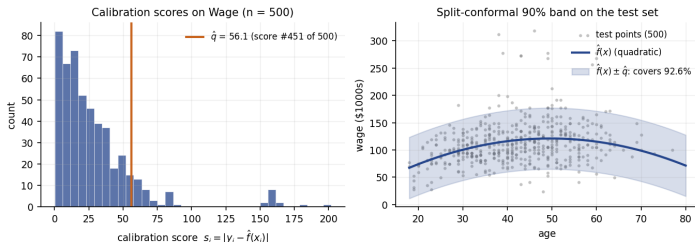
### Proof sketch — three lines of rank arithmetic

- ① Let  $s_{n+1} = |Y_{n+1} - \hat{f}(X_{n+1})|$  join the  $n$  calibration scores. All  $n + 1$  scores are exchangeable (each is the same function of its own data point, with  $\hat{f}$  fixed).
- ② Coverage happens  $\iff s_{n+1} \leq s_{(k)} \iff \text{rank}(s_{n+1}) \leq k$  among the  $n + 1$  scores.
- ③ That rank is uniform on  $\{1, \dots, n + 1\}$ , so coverage  $= \frac{k}{n+1} \geq \frac{(n+1)(1-\alpha)}{n+1} = 1 - \alpha$ . (Full argument: appendix.)

### How tight is the guarantee at $n$ of 500?

$k = 451 \Rightarrow$  coverage is pinned between 0.9 and  $0.9 + \frac{1}{501} = 0.902$  — not just a bound, essentially an *equality*.

# Wage: the calibration quantile and the band it buys



## Takeaway

Quadratic fit on 2000 training points; the 451st of 500 sorted calibration residuals gives  $\hat{q} = 56.1$ . The band  $\hat{f}(\text{age}) \pm 56.1$  covers 92.6% of the 500 test wages — and is *narrower* than the miscalibrated Gaussian band ( $\pm 66.1$ , covering 95.2%).

## Notebook §1: split conformal on Wage

advanced\_03\_conformal\_lab.ipynb, §1 builds exactly this: the split (seed 2024), the sorted scores,  $\hat{q} = 56.14$  and the test coverage 0.926 — about ten lines of numpy.

## Exercise A3.2 — Computing the conformal quantile by hand [Math]

### Exercise

A model is calibrated on  $n = 9$  held-out points with absolute residuals

2.1, 0.7, 4.0, 1.3, 0.5, 2.9, 1.8, 5.2, 1.1.

- 1 Compute  $\hat{q}$  for  $\alpha = 0.2$  and write down  $C(x)$  if  $\hat{f}(x) = 100$ .
- 2 Compute  $\hat{q}$  for  $\alpha = 0.1$ . What do you notice?
- 3 For  $n = 99$  and  $\alpha = 0.05$ : which order statistic is  $\hat{q}$ ?
- 4 Exactly which coverage does the theorem guarantee in part 3 (both bounds)?

# Solution — Exercise A3.2

## Solution

**Method.** Sort, compute  $k = \lceil (n + 1)(1 - \alpha) \rceil$ , pick  $s_{(k)}$ . Sorted scores: 0.5, 0.7, 1.1, 1.3, 1.8, 2.1, 2.9, 4.0, 5.2.

- **Part 1.**  $k = \lceil 10 \times 0.8 \rceil = 8 \Rightarrow \hat{q} = s_{(8)} = 4.0$ , so  $C(x) = [96.0, 104.0]$ .
- **Part 2.**  $k = \lceil 10 \times 0.9 \rceil = 9 \Rightarrow \hat{q} = s_{(9)} = 5.2$  — the *maximum*. With only 9 calibration points, a 90% guarantee forces the most conservative choice available.
- **Part 3.**  $k = \lceil 100 \times 0.95 \rceil = 95 \Rightarrow \hat{q}$  is the **95th smallest of 99** scores.
- **Part 4.** Coverage lies in  $\left[\frac{95}{100}, \frac{95}{100} + \frac{1}{100}\right] = [0.95, 0.96]$  — read off as  $k/(n + 1) = 95/100$  and  $k/(n + 1) + 1/(n + 1)$ .

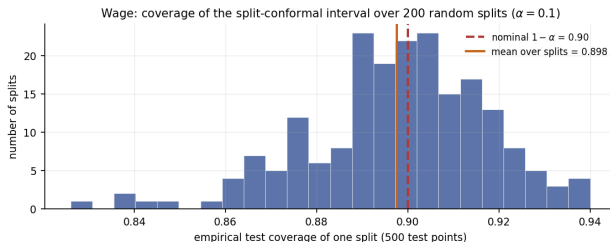
**Take-away.**  $\hat{q}$  is always a *single order statistic* — no interpolation, no model, just a sorted list and one index.

## Common mistake

Computing the plain 90% empirical quantile ( $0.9 \times 9 = 8.1$ , so interpolating near  $s_{(8)}$ ). The conformal index uses  $n + 1$ , not  $n$  — forgetting the  $+1$  is precisely what loses the finite-sample guarantee.



# The guarantee is an average over splits — see it hold



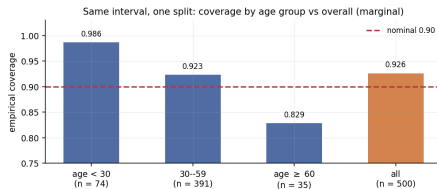
## Takeaway

200 random splits of Wage: per-split test coverage fluctuates between 0.826 and 0.940 (sd 0.020), but the *mean* is  $0.898 \approx 0.9$  — the marginal guarantee in action. Any *single* split can land at 0.87; nothing is wrong when it does.

## Notebook §2: coverage over repeated splits

§2 re-runs the split 200 times and reproduces this histogram — mean 0.8976, sd 0.0198. The appendix derives why the spread is exactly of this size.

# The honest slide: marginal is not conditional



## Takeaway

Same interval, same split: coverage is 98.6% for under-30s but only 82.9% for the 60-plus group — the marginal 92.6% is an *average* that hides both. A constant-width band overcovers where wages are predictable and undercovers where they spread out.

## Common mistake

Promising a customer segment “90% coverage” because the overall model is conformal. Per-subgroup coverage is **not** guaranteed — if a group matters, calibrate on that group (Mondrian conformal, appendix) or use adaptive scores (next section).

## Exercise A3.3 — How small can the calibration set be? [Math]

### Exercise

The conformal index is  $k = \lceil (n+1)(1-\alpha) \rceil$ , and  $\hat{q} = s_{(k)}$  only exists if  $k \leq n$ .

- 1 Show that  $k \leq n$  is equivalent to  $n \geq \frac{1-\alpha}{\alpha} = \frac{1}{\alpha} - 1$ .
- 2 Evaluate the minimal  $n$  for  $\alpha = 0.1$ ,  $0.05$ , and  $0.01$ .
- 3 A team calibrates with  $n = 10$  points at  $\alpha = 0.05$ . What interval does the recipe return, and why is that the honest answer?
- 4 Which coverage levels are *exactly* achievable with  $n = 9$ ?

# Solution — Exercise A3.3

## Solution

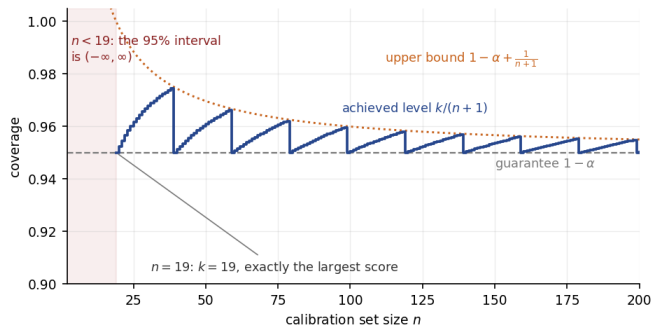
**Method.** Unpack the ceiling:  $k \leq n \iff (n+1)(1-\alpha) \leq n$ .

- **Part 1.**  $(n+1)(1-\alpha) \leq n \iff n+1-\alpha n-\alpha \leq n \iff 1-\alpha \leq \alpha n \iff n \geq \frac{1-\alpha}{\alpha} = \frac{1}{\alpha} - 1$ . ✓
- **Part 2.**  $\alpha = 0.1$ :  $n \geq 9$ .     $\alpha = 0.05$ :  $n \geq 19$ .     $\alpha = 0.01$ :  $n \geq 99$ .
- **Part 3.**  $k = \lceil 11 \times 0.95 \rceil = \lceil 10.45 \rceil = 11 > 10$ : there is no 11th score, so  $\hat{q} = \infty$  and  $C(x) = (-\infty, \infty)$ . Honest, because with 10 exchangeable scores the new point exceeds *all* of them with probability  $1/11 \approx 9.1\% > 5\%$  — no finite cut-off can promise 95%.
- **Part 4.** Achievable exact levels are  $\frac{k}{n+1} = \frac{k}{10}$ : 10%, 20%, ..., 90%, 100% — coverage is *quantized* in steps of  $\frac{1}{n+1}$ .

## Common mistake

Requesting  $\alpha = 0.01$  from  $n = 50$  calibration points and “fixing” the infinite interval by switching to the plain empirical maximum. The interval is then no longer guaranteed — collect more calibration data instead.

# What the calibration set buys, drawn



## Takeaway

Exercise A3.3 as a picture. Below  $n = 19$  the 95% interval is  $(-\infty, \infty)$ ; at exactly  $n = 19$  the quantile is the largest score; and the overshoot above  $1 - \alpha$  decays like  $1/(n+1)$  — the sawtooth is the ceiling function doing its work.

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# One width for everyone is a blunt instrument

The absolute-residual band on Wage is  $\pm 56.1$  *everywhere*: for 20-year-olds (tightly clustered wages) and for 60-year-olds (wide spread) alike.

## Two ways to make the width follow the data

- **Locally weighted scores**: fit a second model  $\hat{\sigma}(x)$  for the typical error size, and score

$$s_i = \frac{|y_i - \hat{f}(x_i)|}{\hat{\sigma}(x_i)} \implies C(x) = \hat{f}(x) \pm \hat{q} \hat{\sigma}(x).$$

- **Conformalized quantile regression (CQR)**: model the *quantiles* of  $Y \mid X$  directly, then calibrate them.

Both keep the marginal guarantee — any fixed score function works, because the scores of exchangeable points remain exchangeable.

## Locally weighted conformal on Wage

$\hat{\sigma}(x)$ : quadratic fit of  $|\text{residual}|$  on age. Result:  $\hat{q} = 1.90$  (in  $\sigma$ -units), test coverage 89.6%, and widths now range from 42 (age 18) to 121 (age 50) instead of a flat 112.

# The engine: quantile regression with the pinball loss

## Pinball (quantile) loss at level $\tau$

$$\rho_{\tau}(u) = \begin{cases} \tau u & u \geq 0 \\ (\tau - 1) u & u < 0 \end{cases} \quad u = y - q.$$

Minimising  $E \rho_{\tau}(Y - q(X))$  over functions  $q$  yields the conditional  $\tau$ -quantile of  $Y | X$ .

## Why it works — in one sentence

Undershooting costs  $\tau$  per unit, overshooting costs  $1 - \tau$ : the optimal  $q(x)$  sits where a fraction  $\tau$  of the mass lies below — exactly the quantile. At  $\tau = 0.5$  it is the absolute loss, and we recover the median. Below,  $\alpha$  is  $\tau$ : the two fits trace the 5% and 95% edges of  $Y | X$  — the raw band that CQR then calibrates.

```
from sklearn.ensemble import GradientBoostingRegressor # Ch. 8 engine
lo = GradientBoostingRegressor(loss="quantile", alpha=0.05, # 5% curve
    n_estimators=200, max_depth=2, learning_rate=0.05) # Ch. 8 knobs
hi = GradientBoostingRegressor(loss="quantile", alpha=0.95, # 95% curve
    n_estimators=200, max_depth=2, learning_rate=0.05) # same knobs
```



# Conformalized quantile regression (CQR)

## Algorithm (Romano, Patterson & Candès, 2019)

- 1 On  $\mathcal{D}_{\text{tr}}$ , fit quantile curves  $\hat{q}_{\text{lo}}$  (level  $\alpha/2$ ) and  $\hat{q}_{\text{hi}}$  (level  $1 - \alpha/2$ ).
- 2 On  $\mathcal{D}_{\text{cal}}$ , score  $s_i = \max(\hat{q}_{\text{lo}}(x_i) - y_i, y_i - \hat{q}_{\text{hi}}(x_i))$
- 3  $\hat{q} = s_{(k)}$  with the usual  $k = \lceil (n+1)(1-\alpha) \rceil$ ; report  $C(x) = [\hat{q}_{\text{lo}}(x) - \hat{q}, \hat{q}_{\text{hi}}(x) + \hat{q}]$ .

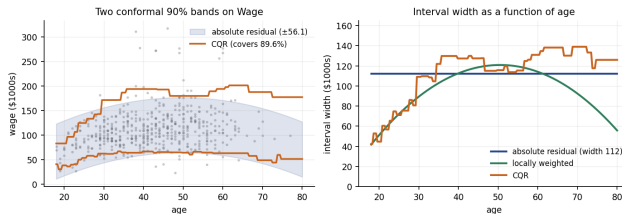
### Decoding the score

$s_i$  is the *signed distance to the nearer violated edge*: positive if  $y_i$  falls outside  $[\hat{q}_{\text{lo}}, \hat{q}_{\text{hi}}]$  (by how much), *negative* if inside (how deep). If the fitted band already overcovers,  $\hat{q} < 0$  *shrinks* it; if it undercovers,  $\hat{q} > 0$  pads it. Same quantile index, same guarantee — the width now lives in the quantile curves.

### Takeaway

CQR = “let Chapter 8 draw the band, let conformal calibrate it”. On Wage the correction is tiny:  $\hat{q} = 0.66$  — the boosted 5%/95% curves were almost calibrated already.

# Wage: constant width vs adaptive width



## Takeaway

Both bands cover  $\approx 90\%$  marginally (CQR: 89.6%), but CQR spends its width where the data spread: average width 70.5 for under-30s vs 134.9 for 60-plus (constant band: 112.3 for both). Note the locally weighted curve *falls* after age 60 — its quadratic  $\hat{\sigma}$  extrapolates poorly: the allocation is only as good as the scale model, even though marginal coverage (89.6%) is untouched.

## Notebook §3: adaptive intervals

§3 fits both adaptive variants and reproduces this figure's numbers:  $\hat{q}_{lw} = 1.897$ ,  $\hat{q}_{cqr} = 0.66$ , and the three width profiles.

## Extended Exercise A3.1 — CQR on Wage, end to end [Python]

### Extended exercise (15 min)

Reproduce the adaptive band. Using the deck's split of Wage (seed 2024; 2000 / 500 / 500):

- 1 Fit `GradientBoostingRegressor(loss="quantile")` at levels 0.05 and 0.95 on the training set (200 trees, depth 2, learning rate 0.05, `random_state=0`).
- 2 Compute the CQR scores on the calibration set and  $\hat{q}$  at  $\alpha = 0.1$ ; report test coverage.
- 3 Compare average interval width for ages  $< 30$ ,  $30\text{--}59$  and  $\geq 60$  against the constant-width band (112.3).
- 4 Compare the two bands' *subgroup coverage* for under-30s and 60-plus. Which failure of the constant band does CQR repair, and what does it *not* guarantee?

### Run this live

The worked solution sits in `advanced_03_conformal_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Extended Exercise A3.1 — CQR on Wage, end to end [Python]*.

## Solution — Extended Exercise A3.1 (1/2)

## Solution — code: fit / score / calibrate

```
import numpy as np, pandas as pd # data handling
from sklearn.ensemble import GradientBoostingRegressor # the CQR engine

wage = pd.read_csv("Wage.csv") # lab: via load()
x, y = wage["age"].to_numpy(float), wage["wage"].to_numpy() # x = age
rng = np.random.default_rng(2024) # the deck's split
i = rng.permutation(len(y)); tr, ca, te = i[:2000], i[2000:2500], i[2500:]
X = x.reshape(-1, 1) # sklearn wants 2-D
gb = dict(n_estimators=200, max_depth=2, learning_rate=0.05, random_state=0)
lo = GradientBoostingRegressor(loss="quantile", alpha=0.05, **gb).fit(X[tr], y[tr])
hi = GradientBoostingRegressor(loss="quantile", alpha=0.95, **gb).fit(X[tr], y[tr])
s = np.maximum(lo.predict(X[ca]) - y[ca], y[ca] - hi.predict(X[ca])) # CQR score
k = int(np.ceil((len(ca) + 1) * 0.9)) # k = 451 of 500
qhat = np.sort(s)[k - 1] # 0.66: tiny correction
L, U = lo.predict(X[te]) - qhat, hi.predict(X[te]) + qhat # pad both edges
print(round(qhat, 2), ((y[te] >= L) & (y[te] <= U)).mean()) # 0.66, cov 0.896
```

# Solution — Extended Exercise A3.1 (2/2)

## Solution — numbers and interpretation

Parts 2–3.  $\hat{q} = 0.66$ ; test coverage = 0.896. Average widths:

|                               | age < 30 | age 30–59 | age $\geq$ 60 |
|-------------------------------|----------|-----------|---------------|
| Constant band (abs. residual) | 112.3    | 112.3     | 112.3         |
| CQR band                      | 70.5     | 120.4     | 134.9         |

**Part 4.** Subgroup coverage on this split: constant band 0.986 (under-30) vs 0.829 (60-plus); CQR 0.797 vs 0.886. CQR repairs the *width misallocation* — it stops wasting 40 units of width on the young and reinvests them in the old, whose coverage rises from 0.829 to 0.886. **What it does not guarantee:** exact per-group coverage — the young bin ( $n = 74$ ) lands at 0.797 here, within sampling noise of 0.9 but not pinned to it. Only the *marginal* 90% is guaranteed; per-group guarantees need per-group calibration.

## Common mistake

Fitting  $\hat{q}_{lo}, \hat{q}_{hi}$  on training *plus* calibration data “to use all the data”. The scores are then not exchangeable with the test score, and the guarantee is void — the split is load-bearing.

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# From intervals to prediction sets

For a classifier there is no interval — the honest object is a **set of classes**  $C(x) \subseteq \{1, \dots, K\}$ .

## Split conformal for classification

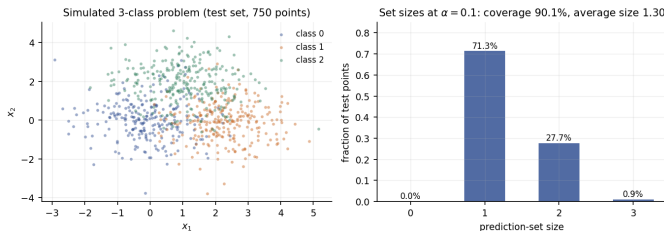
- 1 Fit a probabilistic classifier  $\hat{p}$  on  $\mathcal{D}_{\text{tr}}$  (e.g. softmax / logistic regression, Ch. 4).
- 2 Calibration scores:  $s_i = 1 - \hat{p}_{y_i}(x_i)$  — one minus the probability assigned to the *true* class.
- 3  $\hat{q} = s_{(k)}$ ,  $k = \lceil (n+1)(1-\alpha) \rceil$  as always; report

$$C(x) = \{ k : \hat{p}_k(x) \geq 1 - \hat{q} \}.$$

## Decoding the set rule

A class enters the set when its softmax probability clears the threshold  $1 - \hat{q}$ . Confident points get **singletons**; ambiguous points near a decision boundary get  $\{1, 2\}$  or bigger. The guarantee: the *true* class is in  $C(X)$  with probability  $\geq 1 - \alpha$  — same rank argument, unchanged.

# A 3-class simulation: sets earn their keep at the boundaries



## Takeaway

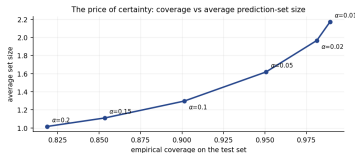
Three Gaussian classes (centres  $(0, 0)$ ,  $(2.2, 0)$ ,  $(1.1, 1.9)$ ; logistic accuracy 80.9%). At  $\alpha = 0.1$ :  $\hat{q} = 0.7685$  (score 676 of 750), coverage 90.1%, average set size 1.30 — 71.3% singletons, 27.7% pairs, 0.9% full sets. The set size is a per-point difficulty meter the accuracy number hides.

## Notebook §4: conformal classification

§4 simulates these three classes (seed 2024), builds the sets and reproduces coverage 0.9013 and the size distribution above.



# The price of certainty: coverage vs set size



## Takeaway

Demanding 99% coverage inflates the average set from 1.30 to 2.17 classes — two-thirds of the label space. Certainty is bought with set size; pick  $\alpha$  by what a large set *costs downstream* (a manual review, a second lab test).

## One slide on APS (adaptive prediction sets)

The  $1 - \hat{p}$  score judges each class alone, so it can return *empty* sets when  $\alpha$  is loose (here: 0.9% empty at  $\alpha = 0.2$ ) and adapts poorly across easy/hard regions. **APS** scores the *cumulative* probability mass needed to reach the true class (add classes in decreasing  $\hat{p}_k$ ); sets are never empty and track difficulty better, at the price of being larger on average. Same calibration machinery.

## Exercise A3.4 — Reading prediction sets [Concept]

### Exercise

A conformal classifier ( $K = 3$ ,  $\alpha = 0.1$ ) was calibrated with  $\hat{q} = 0.7685$ , so classes enter the set when  $\hat{p}_k(x) \geq 0.2315$ . Three test customers get softmax probabilities:

A: (0.95, 0.03, 0.02)      B: (0.50, 0.45, 0.05)      C: (0.40, 0.35, 0.25)

- 1 Write down the prediction set of each customer.
- 2 The plain classifier predicts class 0 for all three. What extra information do the sets carry?
- 3 Can the guarantee be read as “90% of *singleton* sets are correct”? Why not?
- 4 Under which circumstances can the rule return an *empty* set, and what should a production system do with one?

# Solution — Exercise A3.4

## Solution

**Method.** Threshold each probability at  $1 - \hat{q} = 0.2315$ .

- **Part 1.** A:  $\{0\}$  (0.95 clears, others do not). B:  $\{0, 1\}$  (0.50 and 0.45 clear). C:  $\{0, 1, 2\}$  (all three  $\geq 0.2315$ ).
- **Part 2.** Identical point predictions, very different epistemic states: A is settled, B is a genuine two-way toss-up, C is anyone's guess. The set size is the honest “how sure are we” channel a point prediction lacks.
- **Part 3.** No. The guarantee is *marginal over all points*:  $\geq 90\%$  of *all* sets contain the truth. Singletons are the *easy points*, so their hit rate is typically above 90%, while the large sets carry more of the mistakes — but neither is individually guaranteed.
- **Part 4.** Empty sets occur when even the largest probability misses the threshold: possible once  $\hat{q} < 1 - \max_k \hat{p}_k(x)$ , i.e. with loose  $\alpha$  and diffident models (our simulation: 0.9% empty at  $\alpha = 0.2$ , none at  $\alpha = 0.1$ ). Production systems treat empty (like oversized) sets as **abstentions**: route to a human.

## Common mistake

Treating the set as a ranking.  $C(x)$  is an *unordered* guarantee region — “class 0 first” is the old point prediction sneaking back in.

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# Same data, two intervals: when the model is right, they agree

## Homoskedastic simulation — the friendly case

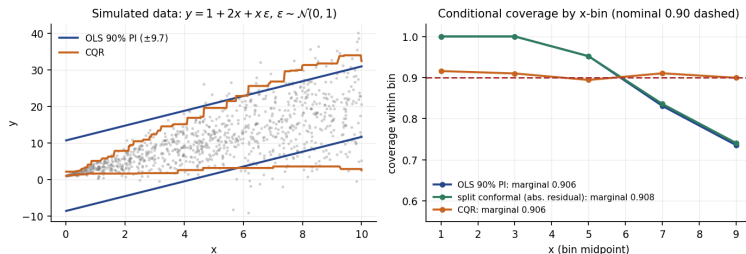
$y = 1 + 2x + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, 2^2)$ ,  $x \sim U(0, 10)$ ; 2 000 training, 1 000 calibration, 1 000 test points (seed 2024). Theory: 90% half-width =  $1.645 \times 2 = 3.29$ .

|                                 | half-width | test coverage |
|---------------------------------|------------|---------------|
| OLS prediction interval (Ch. 3) | 3.31       | 0.915         |
| Split conformal (abs. residual) | 3.24       | 0.909         |

## Takeaway

When the Gaussian linear model is *true*, conformal costs essentially nothing: both intervals reproduce the theoretical  $\pm 3.29$ . Conformal is not a better interval — it is the same interval with a *weaker warranty requirement*. That makes the pair a cheap diagnostic: run both, and if they disagree materially, the *model assumptions* are the prime suspect.

# Heteroskedastic truth: where the 90% quietly stops being 90%



## Takeaway

$y = 1 + 2x + x\varepsilon$ : noise grows with  $x$ . All three methods hold  $\approx 90\%$  *marginally* (OLS 0.906, conformal 0.908, CQR 0.906) — but conditionally the constant-width pair collapses to 73.5% (74.1%) for  $x > 8$  while overcovering at 100% for  $x < 2$ . Only CQR stays near 0.90 in *every* bin (0.916 to 0.899). And only conformal *guarantees* its marginal number: over 100 repeated simulations, OLS averages 0.894 (luck, no warranty), conformal 0.900 (by theorem).

# The failure that actually happens: plug-in intervals from a flexible model

## Same heteroskedastic data — boosted regression

Fit a 500-tree gradient boosting model, then do what practitioners actually do:  $\hat{f}(x) \pm 1.645 \times \hat{\sigma}_{\text{in}}$  with  $\hat{\sigma}_{\text{in}}$  the sd of the *training* residuals.

|                              | $\hat{\sigma}$ used | claimed     | actual test coverage |
|------------------------------|---------------------|-------------|----------------------|
| Plug-in (training residuals) | 3.92                | 90%         | 77.6%                |
| Split conformal (same model) | $\hat{q} = 10.81$   | $\geq 90\%$ | 90.7%                |

The truth: held-out residual sd is 6.12 — the model memorised a third of the noise, and the plug-in interval inherited the optimism.

## Common mistake

Calibrating interval width on data the model has seen. Training residuals of a flexible model are systematically too small; every per-cent of memorised noise is a per-cent of silent undercoverage. The conformal split exists precisely to make this mistake impossible.

## Chapter 3's interval vs split conformal — the comparison slide

|                        | OLS prediction interval                                                                                              | Split conformal                                                                           |
|------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Assumes                | linear mean, Gaussian, homoskedastic errors                                                                          | exchangeability + a clean split                                                           |
| Guarantee              | exact at every $x$ , <i>if</i> the model is true                                                                     | marginal $\geq 1 - \alpha$ , finite-sample, <i>whatever</i> the model                     |
| Works with<br>Width    | the linear model it was derived for<br>$\approx$ constant ( $\hat{\sigma}$ -driven; slightly wider at high leverage) | any $\hat{f}$ : boosting, nets, black boxes<br>constant (abs. residual) or adaptive (CQR) |
| Cost                   | none                                                                                                                 | holds out $n$ calibration points                                                          |
| Under misspecification | silently wrong coverage                                                                                              | coverage intact; only width reacts                                                        |

## Takeaway

Use the OLS interval when you *believe* the Gaussian linear model and want conditional statements. Use conformal when the model is flexible, the errors are ugly, or the coverage number will be *audited*.



## Extended Exercise A3.2 — Stress-testing the two intervals [Integrative]

### Extended exercise (15 min)

Simulate  $y = 1 + 2x + x\varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, 1)$ ,  $x \sim U(0, 10)$ ,  $n = 4\,000$  (split 2 000 / 1 000 / 1 000, seed 2024).

- 1 Fit OLS on the training set; build the 90% prediction interval  $\hat{y} \pm 1.645 \hat{\sigma}$  and measure test coverage overall, for  $x < 2$ , and for  $x > 8$ .
- 2 Build the split-conformal interval (absolute residuals,  $\alpha = 0.1$ ) and measure the same three coverages. What does conformal fix here — and what does it not fix?
- 3 Fit a 500-tree GradientBoostingRegressor; report coverage of  $\hat{f} \pm 1.645 \times \text{sd}(\text{training residuals})$ , then conformalise the same model. Explain the mechanism behind the gap.
- 4 Conclude: which of the three failure modes (skewed errors, heteroskedastic errors, optimistic  $\hat{\sigma}$ ) does split conformal repair marginally, and which needs CQR?

### Run this live

Worked solution in `advanced_03_conformal_lab.ipynb` under *Lecture exercises* — *worked Python solutions*, cell *Extended Exercise A3.2*.

# Solution — Extended Exercise A3.2 (1/2)

## Solution — code: the three intervals

```
import numpy as np, statsmodels.api as sm # OLS and arrays
from sklearn.ensemble import GradientBoostingRegressor # the flexible fit
rng = np.random.default_rng(2024) # the deck's seed
N = 4000; x = rng.uniform(0, 10, N); y = 1 + 2*x + x*rng.normal(0, 1, N)
tr, ca, te = np.arange(2000), np.arange(2000, 3000), np.arange(3000, 4000)
ols = sm.OLS(y[tr], sm.add_constant(x[tr])).fit() # 1. classical PI
half = 1.645 * np.sqrt(ols.scale) # pooled sigma: 5.87
pred = ols.params[0] + ols.params[1] * x[te] # test fitted values
cov = lambda hit: [hit.mean(), hit[x[te] < 2].mean(), hit[x[te] > 8].mean()]
print(cov(np.abs(y[te] - pred) <= half)) # [0.906, 1.000, 0.735]
k = int(np.ceil(1001 * 0.9)) # 2. conformal: k = 901
q = np.sort(np.abs(y[ca] - (ols.params[0] + ols.params[1]*x[ca]))) [k-1]
print(q, cov(np.abs(y[te] - pred) <= q)) # 9.8 [0.908, 1.0, 0.741]
f = GradientBoostingRegressor(n_estimators=500, random_state=0) # 3. plug-in
f.fit(x[tr, None], y[tr]); sig = np.std(y[tr] - f.predict(x[tr, None])) # 3.92
print(cov(np.abs(y[te] - f.predict(x[te, None])) <= 1.645 * sig)) # [0.776, ...]
qg = np.sort(np.abs(y[ca] - f.predict(x[ca, None]))) [k-1] # 10.81
print(cov(np.abs(y[te] - f.predict(x[te, None])) <= qg)) # [0.907, ...]
```

**Output:** cov returns [all,  $x < 2$ ,  $x > 8$ ] — marginal coverage first, then the two conditional ones.

## Solution — Extended Exercise A3.2 (2/2)

### Solution — interpretation

**Part 1.** OLS: marginal 0.906, but 100% for  $x < 2$  and 73.5% for  $x > 8$  — the pooled  $\hat{\sigma} = 5.87$  is one compromise width for noise that ranges from  $\approx 0$  to 10. **Part 2.** Conformal ( $\hat{q} = 9.79$ ): marginal 0.908, subgroups 1.00 / 0.741 — conformal *guarantees the marginal number* (over repeated data: 0.900 vs OLS's drifting 0.894) but a constant-width score cannot repair the *conditional* imbalance. That is CQR's job (it reaches 0.899 at  $x > 8$ ). **Part 3.** Plug-in GBR: claimed 90%, actual 77.6% — training-residual sd (3.92) understates held-out error sd (6.12) because the 500-tree model partly memorised the training noise. Conformalising the *same* model restores 90.7% by pricing the width on unseen data ( $\hat{q} = 10.81$ ). **Part 4.** Marginal repairs come free (skew, heteroskedasticity, optimism all handled — the guarantee never moved); *conditional* honesty across  $x$  is the one thing that needs an adaptive score (CQR / locally weighted).

### Common mistake

Concluding from part 2 that conformal “failed”. It delivered exactly what it promises — marginal coverage; expecting per-region coverage from a constant-width score is a specification error, not a method error.

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# Pitfall 1: exchangeability violations in the wild

- **Time series:** calibration residuals from a five-year window, deployment two years later — trends, seasonality and regime shifts make the test point *special*, which is exactly what exchangeability forbids.
- **Covariate shift:** a marketing model calibrated on organic traffic, deployed during a paid campaign — the  $X$ -mix changed overnight.
- **Selection feedback:** the model approves loans, and only approved loans generate future calibration labels.

## What to do instead

Time series: calibrate on the *most recent* block, recalibrate on a schedule, or use online/adaptive conformal that tunes  $\alpha$  from realised coverage. Known covariate shift: *weighted* conformal with likelihood-ratio weights  $w(x) = p_{\text{new}}(x)/p_{\text{cal}}(x)$ . Always: monitor realised coverage — the guarantee you can check is the one you keep. A wind-power desk does exactly this, recalibrating on the trailing 30 days: a fixed calibration set from spring undercovers in autumn storms — same model, dead guarantee.

## Pitfall 2: leakage between fitting and calibration

The guarantee has exactly one hard requirement inside your pipeline: *nothing about  $\hat{f}$  may depend on the calibration set.*

### The four common leak channels

- ❶ **Preprocessing:** scaler / imputer / encoder fit on *all* data before the split.
- ❷ **Tuning:** hyperparameters or early stopping selected on calibration performance.
- ❸ **Feature selection:** screening variables on the full sample (the Ch. 5 classic).
- ❹ **Duplicates:** the same customer's rows on both sides of the split.

Each one shrinks calibration scores below test scores  $\Rightarrow \hat{q}$  too small  $\Rightarrow$  silent undercoverage — the plug-in GBR's 77.6% was this mechanism in its purest form.

### Common mistake

“Refitting on train + calibration after computing  $\hat{q}$ , since the interval is already fixed.” The deployed  $\hat{f}$  is then not the one that generated the scores — the pair  $(\hat{f}, \hat{q})$  is calibrated *together* or not at all.

# From ten lines to production

## Practical sizing rules

- **Feasibility:**  $\alpha$  needs  $n \geq \frac{1}{\alpha} - 1$  (19 points for 95%, 99 for 99%) — below that the honest interval is infinite (Exercise A3.3).
- **Stability:** with  $n = 500$  the coverage of one split has  $\text{sd} \approx 1.3\%$  (appendix);  $n$  between 500 and 1 000 is the usual sweet spot — beyond that, spend the data on training.
- **Report the nature of the guarantee:** marginal, at the stated  $\alpha$ , under exchangeability — three words that pre-empt three misreadings.

## The production tool

Teams reach for **MAPIE** (*Model Agnostic Prediction Interval Estimator*, scikit-learn compatible): split/cross conformal, CQR and classification sets behind one API, plus jackknife+ variants that recycle training folds. It is not installed here — deliberately: everything in this module was ten lines from scratch, and those ten lines are the audit trail you will be asked for.

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# Python lab — run it live in the notebook

## Companion notebook: `advanced_03_conformal_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook and run it cell by cell:

- §1, *Split conformal on Wage* — the ten-line recipe:  $\hat{q} = 56.14$ , coverage 0.926;
- §2, *Coverage over repeated splits* — 200 splits, mean 0.898;
- §3, *Adaptive intervals* — locally weighted scores and CQR, width profiles by age;
- §4, *Conformal classification* — 3-class simulation, prediction sets, size distribution;
- §5, *OLS prediction interval vs conformal* — the friendly homoskedastic case (the heteroskedastic stress test is worked as Extended Exercise A3.2).

Data loads via the ISLP package or the bundled CSV files; §6, *Exercises* ends the notebook with hands-on tasks.

# Split conformal in ten lines

```
import numpy as np, pandas as pd # arrays + CSV
from sklearn.linear_model import LinearRegression # any model works

wage = pd.read_csv("Wage.csv") # lab: via load()
x, y = wage["age"].to_numpy(float), wage["wage"].to_numpy() # x = age
rng = np.random.default_rng(2024) # reproducible split
i = rng.permutation(len(y)); tr, ca, te = i[:2000], i[2000:2500], i[2500:]
P = lambda a: np.column_stack([a, a**2]) # quadratic in age
f = LinearRegression().fit(P(x[tr]), y[tr]) # 1. fit on TRAIN only
s = np.abs(y[ca] - f.predict(P(x[ca]))) # 2. calibration scores
k = int(np.ceil((len(ca) + 1) * (1 - 0.1))) # 3. k = ceil((n+1)(1-a))
qhat = np.sort(s)[k - 1] # k-th smallest score
cov = (np.abs(y[te] - f.predict(P(x[te]))) <= qhat).mean() # 4. coverage
print(k, qhat.round(2), cov) # 451 56.14 0.926
```

## How to read this

**Key call:** `np.sort(s)[k-1]`, the 451st smallest held-out residual:  $\hat{q} = 56.14$ , so  $\hat{f}(x) \pm 56.14$  covers 92.6%.

## Exercise A3.5 — Conformal classification from scratch [Python]

### Exercise

Build prediction sets for the deck's 3-class simulation (three Gaussian classes, 1 000 points each, centres  $(0, 0)$ ,  $(2.2, 0)$ ,  $(1.1, 1.9)$ , unit variance, seed 2024; split 1 500 / 750 / 750).

- 1 Fit `LogisticRegression` on the training set.
- 2 Compute the calibration scores  $s_i = 1 - \hat{p}_{y_i}(x_i)$  and  $\hat{q}$  at  $\alpha = 0.1$  (which  $k$ ?).
- 3 Build the test-set prediction sets and report coverage, average set size, and the share of singleton sets.
- 4 Re-run with  $\alpha = 0.01$ : what happens to the average set size?

### Run this live

Worked solution in `advanced_03_conformal_lab.ipynb` under *Lecture exercises — worked Python solutions*, cell *Exercise A3.5*.

# Solution — Exercise A3.5

## Solution — code and expected numbers

```
import numpy as np # arrays only
from sklearn.linear_model import LogisticRegression # Ch. 4 model
rng = np.random.default_rng(2024) # deck's seed
mu = np.array([[0, 0], [2.2, 0], [1.1, 1.9]]) # class centres
X = np.vstack([rng.normal(m, 1.0, (1000, 2)) for m in mu]) # 1000 each
y = np.repeat([0, 1, 2], 1000) # labels, in order
p = rng.permutation(3000); X, y = X[p], y[p] # shuffle once
clf = LogisticRegression(max_iter=1000).fit(X[:1500], y[:1500]) # TRAIN
pc = clf.predict_proba(X[1500:2250]) # calibration probs
s = 1 - pc[np.arange(750), y[1500:2250]] # score: 1 - p_true
k = int(np.ceil(751 * 0.9)) # k = 676 of 750
qhat = np.sort(s)[k - 1] # qhat = 0.7685
sets = clf.predict_proba(X[2250:]) >= 1 - qhat # keep p >= 0.2315
cov = sets[np.arange(750), y[2250:]].mean() # truth in the set?
print(qhat.round(4), cov, sets.sum(1).mean().round(3)) # 0.7685 0.9013 1.296
```

**Parts 3–4.** Coverage 0.9013, average size 1.30, singletons 71.3%. At  $\alpha = 0.01$ :  $k = \lceil 751 \times 0.99 \rceil = 744$ ,  $\hat{q} = 0.9638$ , coverage 98.9% — and the average set grows to 2.17 of 3 classes.

## Common mistake

Scoring with  $1 - \max_k \hat{p}_k$  (the *predicted* class) instead of  $1 - \hat{p}_{y_i}$  (the *true* class). The score must know the truth on the calibration set — that is what it is for.

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# Module A3 in one slide

## What we accomplished

Turned any point predictor from this course into a set-valued predictor with a finite-sample coverage guarantee.

- Exchangeability — the single assumption, and what breaks it.
- Split conformal: scores,  $k = \lceil (n+1)(1-\alpha) \rceil$ , interval  $\hat{f}(x) \pm \hat{q}$ ; proof by rank arithmetic.
- The guarantee is *marginal* — averaged over points and splits, not per subgroup.
- Adaptive widths: locally weighted scores and conformalized quantile regression.
- Classification: prediction sets, set size as a difficulty meter, APS.
- Versus Chapter 3: agreement under the Gaussian model, guarantees under misspecification, and the plug-in trap.

# Key formulas at a glance

**Conformal index**  $k = \lceil (n+1)(1-\alpha) \rceil$ ;  $\hat{q} = s_{(k)}$ ; finite iff  $n \geq \frac{1}{\alpha} - 1$ .

**Regression interval**  $s_i = |y_i - \hat{f}(x_i)|$ ;  $C(x) = [\hat{f}(x) - \hat{q}, \hat{f}(x) + \hat{q}]$ .

**Coverage guarantee**  $1 - \alpha \leq \Pr(Y_{n+1} \in C(X_{n+1})) \leq 1 - \alpha + \frac{1}{n+1}$  (distinct scores).

**Locally weighted**  $s_i = |y_i - \hat{f}(x_i)| / \hat{\sigma}(x_i)$ ;  $C(x) = \hat{f}(x) \pm \hat{q} \hat{\sigma}(x)$ .

**CQR**  $s_i = \max(\hat{q}_{lo}(x_i) - y_i, y_i - \hat{q}_{hi}(x_i))$ ;  $C(x) = [\hat{q}_{lo}(x) - \hat{q}, \hat{q}_{hi}(x) + \hat{q}]$ .

**Classification set**  $s_i = 1 - \hat{p}_{y_i}(x_i)$ ;  $C(x) = \{k : \hat{p}_k(x) \geq 1 - \hat{q}\}$ .

**Pinball loss**  $\rho_\tau(u) = \tau u [u \geq 0] + (\tau - 1)u [u < 0]$  — minimised by the  $\tau$ -quantile.

# Vocabulary checklist

| Term                   | Meaning                                                            |
|------------------------|--------------------------------------------------------------------|
| Exchangeability        | every ordering of the data equally likely (weaker than i.i.d.)     |
| Calibration set        | held-out data that prices the interval width                       |
| Conformity score       | how badly a point disagrees with the model's prediction            |
| $\hat{q}$              | the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest calibration score |
| Marginal coverage      | $\geq 1 - \alpha$ on average over points and splits                |
| Conditional coverage   | coverage at a specific $x$ / subgroup — <i>not</i> guaranteed      |
| Locally weighted CP    | residuals scaled by a fitted error-size model                      |
| CQR                    | conformalized quantile regression — calibrated quantile band       |
| Prediction set         | set of classes guaranteed to contain the truth                     |
| APS                    | adaptive prediction sets via cumulative probability mass           |
| Weighted / Mondrian CP | repairs for covariate shift / per-group guarantees                 |



# Decision rules of thumb

- Any black-box model, honest uncertainty needed  $\Rightarrow$  **split conformal** — ten lines, one held-out slice.
- Spread visibly varies with  $x \Rightarrow$  **CQR** (or locally weighted scores); constant width wastes coverage on the easy region.
- Classifier feeding a triage / review queue  $\Rightarrow$  **prediction sets**; route big or empty sets to a human.
- Believe the Gaussian linear model and need per- $x$  statements  $\Rightarrow$  Chapter 3's interval is fine — and conformal is a free cross-check.
- Time series or known shift  $\Rightarrow$  recent-block calibration, weighted or online conformal; *never* vanilla split conformal untouched.
- Sizing:  $n_{\text{cal}} \approx 500\text{--}1\,000$ ; feasibility floor  $n \geq 1/\alpha - 1$ .
- A subgroup matters on its own  $\Rightarrow$  calibrate within the subgroup.

# Common pitfalls

## Don't

- ❶ Tune hyperparameters, early stopping, scalers or features on the calibration set — that is leakage, and it undercovers silently.
- ❷ Read “90% marginal” as “90% for every customer segment”.
- ❸ Use training residuals as calibration scores (the 77.6% trap).
- ❹ Apply vanilla split conformal to time series and call it guaranteed.
- ❺ Demand  $\alpha = 0.01$  from 50 calibration points — infeasible by arithmetic.
- ❻ Keep last year's  $\hat{q}$  after retraining the model or after the world drifts — the pair  $(\hat{f}, \hat{q})$  is calibrated together.

# Self-check questions

## Can you answer these without looking back?

- 1 Why  $\lceil (n+1)(1-\alpha) \rceil$  and not the plain  $(1-\alpha)$ -quantile of the  $n$  scores?
- 2 State the two-sided coverage bound. Where does each side come from?
- 3 Your single split shows coverage 0.87 at  $\alpha = 0.1$ . Is the method broken?
- 4 What exactly does CQR's  $\max(\hat{q}_{lo} - y, y - \hat{q}_{hi})$  score measure, and when is  $\hat{q}$  negative?
- 5 Why can a prediction set be empty under the  $1 - \hat{p}$  score, and which score family avoids that?
- 6 Name the four leakage channels between fitting and calibration.

## Where to look

1–2: Split Conformal section (and the appendix proof). 3: the coverage histogram. 4: Adaptive Intervals. 5: the APS slide. 6: Pitfall 2.

## Appendix — what is in here, and why

Nothing in the appendix is needed to follow the module: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

### Contents of the appendix

- **The coverage theorem, proved in full** — the rank argument with both bounds and the tie-breaking detail; the main deck only sketched it.
- **How coverage fluctuates: the Beta law** — the exact distribution of one split's coverage, moved out because it needs the Beta distribution.
- **Full conformal and jackknife+** — what you gain and pay when you refuse to split; a side topic beyond the split recipe.
- **Mondrian (group-conditional) conformal** — the fix for the subgroup problem; deferred until the marginal story was complete.
- **A drill exercise on the Beta law** — connects the histogram of the main deck to the formula here.

# The coverage theorem, proved in full

**Setup.** Scores  $s_1, \dots, s_n$  (calibration) and  $s_{n+1}$  (test) are exchangeable;  $k = \lceil (n+1)(1-\alpha) \rceil$ ; coverage event  $A = \{s_{n+1} \leq s_{(k)}\}$  with  $s_{(k)}$  the  $k$ -th smallest of  $s_1, \dots, s_n$ .

## Step 1 — the event is a rank statement

Assume first no ties. Let  $r$  be the rank of  $s_{n+1}$  among all  $n+1$  scores. Exactly  $r-1$  calibration scores lie below  $s_{n+1}$ , hence  $s_{n+1} \leq s_{(k)} \iff r-1 \leq k-1 \iff r \leq k$ .

## Step 2 — the rank is uniform

Exchangeability makes every assignment of the  $n+1$  scores to the  $n+1$  ranks equally likely, so  $\Pr(r=j) = \frac{1}{n+1}$  for each  $j$  and  $\Pr(A) = \frac{k}{n+1}$ .

## Step 3 — both bounds

Lower:  $k \geq (n+1)(1-\alpha) \Rightarrow \Pr(A) \geq 1-\alpha$ . Upper:  $k < (n+1)(1-\alpha) + 1 \Rightarrow \Pr(A) < 1-\alpha + \frac{1}{n+1}$ .  
**Ties:** with ties, break them by adding i.i.d. uniform jitter to every score — exchangeability is preserved, the jittered interval is (weakly) inside the original one, so the *lower* bound survives; only the upper bound needs the distinct-scores assumption.

# How coverage fluctuates from split to split: the Beta law

## Conditional coverage of one calibration set

Given the calibration set, the interval's true coverage is  $F(\hat{q})$ , where  $F$  is the score distribution. For continuous  $F$ ,  $F(s_{(k)}) \sim \text{Beta}(k, n+1-k)$ , hence

$$E[F(\hat{q})] = \frac{k}{n+1}, \quad \text{sd} = \sqrt{\frac{k(n+1-k)}{(n+1)^2(n+2)}}.$$

## Predicting the Wage histogram before running it

$n = 500$ ,  $k = 451$ : mean =  $\frac{451}{501} = 0.9002$ , sd = 0.0134. One split's *measured* coverage adds test-set noise ( $\sqrt{0.9 \times 0.1/500} = 0.0134$ ); combined:  $\sqrt{0.0134^2 + 0.0134^2} = 0.019$ . Observed over 200 splits: sd = 0.0198. Theory and histogram agree to the third decimal.

## Takeaway

The guarantee fixes the *mean* of the coverage distribution; the Beta law fixes its *spread*. Doubling  $n$  shrinks the sd by  $\sqrt{2}$  — this is the calculation behind the “ $n \approx 500$ –1 000 suffices” rule of thumb.

## Refusing to split: full conformal and jackknife+

| Method          | Idea                                                                                      | Model fits                     | Guarantee          |
|-----------------|-------------------------------------------------------------------------------------------|--------------------------------|--------------------|
| Split conformal | one split, one fit                                                                        | 1                              | $\geq 1 - \alpha$  |
| Full conformal  | for every candidate $y$ , refit including $(x, y)$ and test its score's rank $y$ (a grid) | one per candidate $y$ (a grid) | $\geq 1 - \alpha$  |
| Jackknife+      | $n$ leave-one-out fits; intervals from LOO residuals                                      | $n$                            | $\geq 1 - 2\alpha$ |
| CV+             | $K$ -fold version of jackknife+                                                           | $K$                            | $\geq 1 - 2\alpha$ |

## Why this module taught the split version

Full conformal uses every point for both fitting and calibration — statistically the most efficient, computationally absurd for boosting or neural nets. Jackknife+/CV+ sit in between, with a weaker worst-case constant ( $1 - 2\alpha$ ; in practice close to  $1 - \alpha$ ). Split conformal costs one held-out slice and *one* model fit — which is why it is the production default and the MAPIE default.

# Mondrian conformal: buying back the subgroups

## Group-conditional (Mondrian) conformal

Partition the space into groups  $g(x)$  fixed *in advance* (age bands, product lines, hospitals). Calibrate a separate  $\hat{q}_g$  from the calibration scores of each group. Then, for every group,  $\Pr(Y \in C(X) \mid g(X) = g) \geq 1 - \alpha$ .

### The price

Each group needs its own feasible calibration count ( $n_g \geq \frac{1}{\alpha} - 1$ , and comfortably more for stable widths — the Beta law now applies *per group*). On Wage, the 60-plus group had only 35 test points and  $\sim 50$  calibration points per split at  $\alpha = 0.1$ : feasible, but with visibly noisy widths. Fully *conditional* coverage at every single  $x$  is provably impossible distribution-free — groups are the honest middle ground.

### Takeaway

Marginal coverage is free; group coverage costs calibration data per group; pointwise coverage cannot be bought at any price (without assumptions).



## Appendix Exercise A3.A — The Beta law in your hands [Math]

### Exercise

A team calibrates with  $n = 99$  points at  $\alpha = 0.1$ .

- 1 Compute  $k$  and the exact expected coverage  $k/(n+1)$ .
- 2 Using the Beta law, compute the sd of the true coverage of one calibration draw. (Hint:  $\text{sd} = \sqrt{k(n+1-k)/((n+1)\sqrt{n+2})}$ .)
- 3 The team wants the sd below 0.01. Roughly which  $n$  do they need (use  $\text{sd} \approx \sqrt{\alpha(1-\alpha)/n}$ )?

# Solution — Appendix Exercise A3.A

## Solution

**Method.** Plug into the Beta( $k, n + 1 - k$ ) moments.

- **Part 1.**  $k = \lceil 100 \times 0.9 \rceil = 90$ ; expected coverage  $= \frac{90}{100} = 0.900$  exactly.
- **Part 2.**  $\text{sd} = \sqrt{90 \times 10} / (100\sqrt{101}) = 30/1004.99 \approx 0.030$ : a single calibration draw's true coverage typically sits anywhere in  $\approx 0.87$ – $0.93$ .
- **Part 3.**  $\sqrt{0.09/n} \leq 0.01 \iff n \geq 900$ . Nine hundred calibration points buy percent-level stability — consistent with the deck's 500–1 000 rule of thumb ( $n = 500$  gave sd 0.0134).

*Common mistake:* treating the guarantee's *mean* statement as if it removed this spread. The theorem constrains the average over calibration draws; the Beta law says how far your *one* draw can sit from it.